P14 calibration

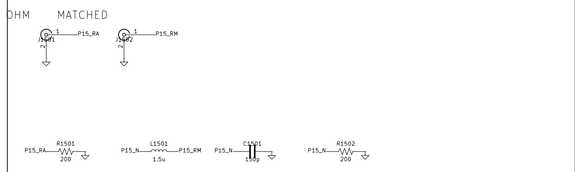
cells (0.0), (1.0), (2.0)



EXPECT: SHORT, OPEN and LOAD at the jacks' pins and a 19.8 mm THRU: a full SOLT calibration; the 25 ohm standard then reads |S11| = 1/3 (9.5 dB return loss)
Calibration standards at the board surface. SHORT: the centre pin poured to ground all round. OPEN: the bare pin (fringing about 30 fF, to be measured against the NanoVNA's SMA open). LOAD: 4 mm of 50 ohm line into 2 × 100 ohm. 25 OHM: 2 × 49.9 ohm at the end of 4.75 mm of line, a verification standard (VSWR 2). THRU: 19.8 mm of 50 ohm line, about 115 ps (enter it as the thru's delay). Every jack on the boards is the same part, so a calibration here measures each tile from its pads with its jacks taken out.

P15 two loads

cell (3.0)



EXPECT: 200 OHM: Gamma = +0.6 (VSWR 4) at every frequency below 100 MHz. MATCHED: Gamma = 0 at 9.2 MHz, |Gamma| 0.5 at about 5 and 14 MHz
An L-network: 1.5 uH in series from the jack, 150 pF and 200 ohm to ground. Z = j ω L = 200 / (1 + j ω 200 C) is 50 ohm where ω 200 C = sqrt(3), 9.2 MHz; away from it the point circles the chart. The 200 ohm alone is the unmatched load.

P16 resistive splitter

cells (0.1), (1.1)



EXPECT: −6.0 dB from IN to each OUT, DC to 6 GHz, every port 50 ohm; OUT 1 to OUT 2 also −6 dB (no isolation)
Three 16.9 ohm (50 / 3) from a star to the three ports: each port sees 16.9 + (16.9 ∥ 50) / 2 = 50 ohm, and each output gets half the input's voltage. Symmetric, so any port is IN. Compare P17's transformer splitter (3 dB, 25 dB of isolation, 0.5–400 MHz).

P17 transformer splitter

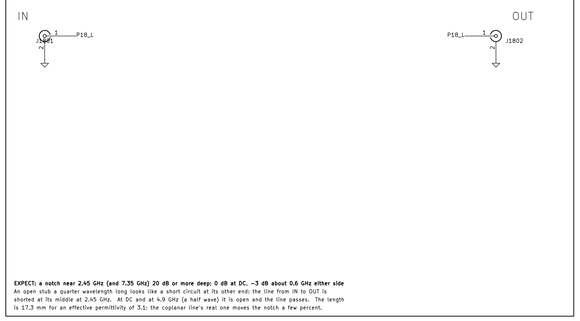
cells (2.1), (3.1)



EXPECT: −3.1 to −3.4 dB from SUM to each port, 0.5–400 MHz; 25–34 dB between port 1 and port 2 (the datasheet)
Mini-Circuits ADP-2-1+: a transformer splitter (a centre-tapped winding with a resistor across its ends), so the two ports are isolated from each other: a combiner for two generators (the two DACs into one amplifier, for a two-tone intermodulation test) as well as a splitter. Pins 2 and 5 are joined and must not be grounded.

P18 quarter-wave stub

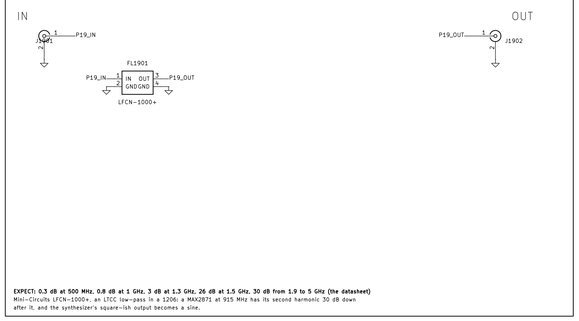
cell (0.2)



EXPECT: a notch near 2.45 GHz (and 7.35 GHz) 20 dB or more deep; 0 dB at DC, −3 dB about 0.6 GHz either side
An open stub a quarter wavelength long looks like a short circuit at its other end: the line from IN to OUT is shorted at its middle at 2.45 GHz. At DC and at 4.9 GHz (a half wave) it is open and the line passes. The length is 17.3 mm for an effective permittivity of 3.1; the coplaner line's real one moves the notch a few percent.

P19 low-pass

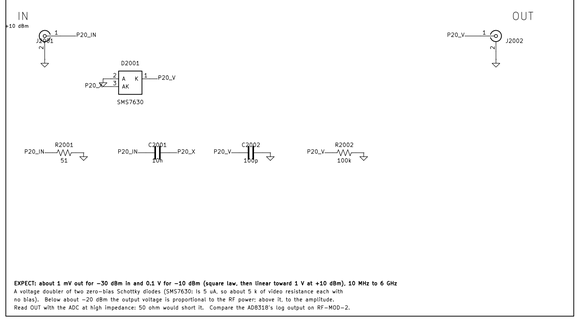
cell (1.2)



EXPECT: 0.3 dB at 500 MHz, 0.8 dB at 1 GHz, 3 dB at 1.3 GHz, 26 dB at 1.5 GHz, 30 dB from 1.9 to 5 GHz (the datasheet)
Mini-Circuits LFCN-1000+, an LTCC low-pass in a 1206; a MAX2871 at 915 MHz has its second harmonic 30 dB down after it, and the synthesizer's square-ish output becomes a sine.

P20 detector

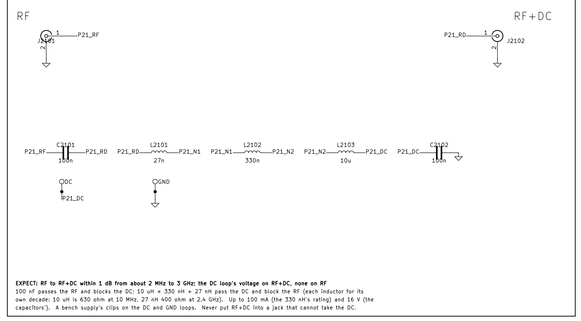
cell (2.2)



EXPECT: about 1 mV out for −30 dBm in and 0.1 V for −10 dBm (square law, then linear toward 1 V at +10 dBm), 10 MHz to 6 GHz
A voltage doubler of two zero-bias Schottky diodes (SMS7630: is 5 uA, so about 5 k of video resistance each with no bias). Below about −20 dBm the output voltage is proportional to the RF power: above it, to the amplitude. Read OUT with the ADC at high impedance; 50 ohm would short it. Compare the AD8318's log output on RF-MOD-2.

P21 bias-T

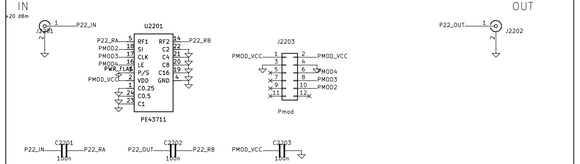
cell (3.2)



EXPECT: RF to RF+DC within 1 dB from about 2 MHz to 3 GHz; the DC loop's voltage on RF+DC, none on RF
100 nF passes the RF and blocks the DC; 10 uH + 330 nH + 27 nH pass the DC and block the RF (each inductor for its own decade; 10 uH is 630 ohm at 10 MHz, 27 nH 400 ohm at 2.4 GHz). Up to 100 mA (the 330 nH's rating) and 16 V (the capacitors). A bench supply's clips on the DC and GND loops. Never put RF+DC into a jack that cannot take the DC.

P22 step attenuator

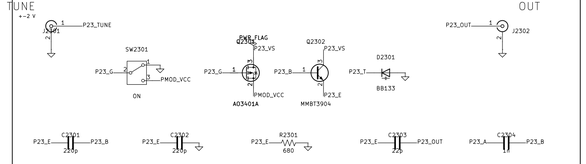
cell (0.3)



EXPECT: 0 to 31.75 dB in 0.25 dB steps (1.5–2.8 dB of loss at 0 dB, rising with frequency), set by an 8-bit word over the Pmod: 31.75 dB at power-up
pSemi PE43711 in serial mode: shift 8 bits (bit 7 = 0, then 4 × dB, least significant first) on SDATA (D2) with SCLK (D3) while LE (D4) is low, then raise LE. RF-MOD's MAX2871 and TRF3722 take the same three lines' timing. Below 48 MHz it takes less power: about +9.5 dBm under 0.5 MHz and +15 dBm at 5 MHz (+23 dBm above 48 MHz).

P23 oscillator

cell (1.3)



EXPECT: about 10.5 MHz with TUNE at 0 V, 9.6 MHz at −2 V and 11.5 MHz at +2 V; about −22 dBm into 50 ohm; locks to a tone on TUNE within a few kHz of it
A Clapp oscillator: the emitter follower's 220 pF / 220 pF divider feeds back what the series LC (10 uH and the BB133 passes at its resonance, f = 1 / (2 pi sqrt(L C)) with C the three capacitors in series. The varactor's cathode is at (5.3 V + TUNE) / 2, so TUNE tunes it (an FM transmitter from a DAC, or the control of a PLL closed in the FPGA): a tone on TUNE near it also couples through 2.2 pF and the oscillator locks to it over a range that grows with the tone's amplitude (Adler's equation). Compare its phase noise with the MAX2871's. The switch turns it off.

P24 1:1 transformer

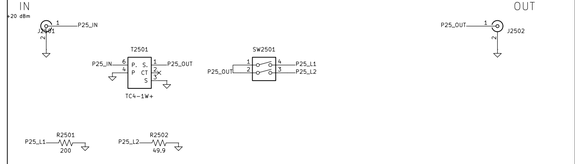
cell (2.3)



EXPECT: 0.3 dB of loss at 10 MHz, 0.5 dB at 100 MHz, 1.8 dB at 500 MHz (the datasheet); OUT's shell is not ground
Mini-Circuits TC1-1TG2+: two windings, so OUT's centre and shell are a balanced pair; a balun for a dipole (an SMA to two-clip lead on OUT), or a break in a ground loop between the host and a mains-earthed instrument. OUT's legs are their own net with no copper of the board's ground within 5.6 mm on any layer.

P25 1:4 transformer

cell (3.3)



EXPECT: IN reads 50 ohm with switch 1 on (200 ohm across OUT) and 12.5 ohm with switch 2: OUT's voltage is twice IN's, 0.8 dB of loss at 10 MHz
Mini-Circuits TC4-1W+: the secondary has twice the primary's turns, so it carries twice the voltage at half the current and any load on OUT looks a quarter of itself at IN. 200 ohm on OUT (switch 1) is a match: 49.9 ohm (switch 2) reads 12.5 ohm. The centre tap is left open.

GND — no mounting holes: four stick-on rubber feet, marked on the back

GND GND GND GND — ground loops for the scope's clips, one per channel, in the board's margins

PWR_FLAG

Drawn by scripts/gen_sch.py from boards/amod_p2.py: the expected responses from sim/passive.py
Every tile is one circuit with its own SMA jacks; the board's silkscreen draws each tile's wires as they
calibration standards, two splitters, a detector, a bias-T, a stub, a low-pass, transformers, a step at
Board: Jason Gallicchio and Claude | open hardware, CERN-OHL-P-2.0 | github.com/gallicchio/v

Sheet: /
File: amod_p2.kicad_sch

Title: AMOD-P2: passive RF and analog blocks

Size: A1	Date: 2026-10-25	Rev: A
KiCad E.D.A. 10.0.7		Id: 1/1